



EUREKA FORBES

Vendor Portal Managment



Bhavesb

NIBHA INFOTECH PRIVATE LIMITED

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1. Business Requirement Document (BRD)

1.1 Objective

To build a Vendor Collaboration Portal integrated with SAP ECC 6.0 EHP8 to enable:

- Seamless PO communication between SAP and vendor
- Vendor collaboration — Acknowledgement and Advance Shipment Notification (ASN)
- Packaging and serial number tracking
- Payment transparency and self-service visibility
- Vendor self-service and delegation

1.2 Stakeholders

Role	Responsibility
Procurement Team	PO creation, ASN approval
Vendors	PO processing, dispatch
Finance Team	Payments
IT Team	Integration & support

1.3 Scope

In Scope

- PO Integration (SAP ME21N to Portal via API / IDoc)
- Vendor Acknowledgement — Accept, Reject, Modify (Quantity & Delivery Date)
- ASN Management
- Box & Serial Number Creation and Mapping
- Label Printing (Outer Box, Inner Box — PDF / Zebra)
- Payment Visibility
- User Management and Delegation
- Mobile Application

Out of Scope

- Supplier onboarding workflows (optional future phase)
- Contract management

2. Functional Specification (FSD)

2.1 Module 1: PO Integration

Process flow:

- PO is created in SAP via transaction ME21N
- PO is sent to the Vendor Portal via API or IDoc
- PO becomes available and visible on the vendor dashboard

2.2 Module 2: Vendor Acknowledgement

Vendor can:

- Accept or Reject a PO
- Modify Quantity and Delivery Date

2.3 Module 3: ASN Management

Vendor:

- Creates ASN with shipment details
- Enters transporter information and dispatch date

Buyer:

- Reviews and approves the ASN before dispatch proceeds

2.4 Module 4: Box & Serial Management

Key Features:

- Create Outer Boxes
- Create Inner Boxes
- Map serial numbers to boxes

Business Rules:

- Serial number must be unique
- Total serials must match PO quantity
- Box hierarchy is mandatory (Outer > Inner > Serial)

2.5 Module 5: Label Printing

Supports printing of:

- Outer Box Labels (QR-code based)
- Inner Box Labels
- Output formats: PDF and Zebra thermal printer



2.6 Module 6: Payment Visibility

Vendor can view:

- Invoice details
- Current payment status
- Full payment history

2.7 Module 7: User Management

Vendor Admin capabilities:

- Create delegation users
- Assign roles
- Add and manage mobile numbers

2.8 Module 8: Security

- Password change and reset workflows
- OTP-based login
- Role-based access control (RBAC)

3. Key Screen Wireframes

3.1 Dashboard

Open POs	Pending Ack	ASN	Payments
View & manage all open purchase orders	Orders awaiting acknowledgement	Shipment notifications	Invoice & payment status

3.2 PO Screen

Fields: PO Number | Material | Quantity | Editable Quantity | Delivery Date

Actions: [Accept] [Reject] [Modify & Submit]

3.3 ASN Screen

Fields: Shipment Date | Transporter Name | Vehicle Number

Actions: [Create ASN] [Submit for Approval]

3.4 Box Creation Screen

Hierarchy: Outer Box → Inner Box → Serial Number Mapping

Actions: [Add Box] [Map Serials] [Validate]

3.5 Payment Screen

Invoice No.	Amount	Status	Due Date
INV-2024-001	₹1,25,000	Paid	15 Jan 2025
INV-2024-002	₹87,500	Pending	28 Feb 2025

4. API Specification

4.1 SAP → Portal

Method	Endpoint	Description
POST	/api/po/create	Receive new PO from SAP and publish to vendor dashboard

4.2 Portal → SAP

Method	Endpoint	Description
POST	/api/po/ack	Send vendor acknowledgement back to SAP
POST	/api/asn/create	Create Advance Shipment Notification with transporter details
POST	/api/serial/upload	Upload and validate serial numbers against PO quantity

4.3 Payment APIs

Method	Endpoint	Description
GET	/api/payment/status	Fetch current payment status for an invoice
GET	/api/payment/history	Fetch full payment history for a vendor

4.4 User APIs

Method	Endpoint	Description
POST	/api/user/create	Create delegation user with role and mobile assignment
POST	/api/user/password/change	Change or reset user password with OTP validation

5. End-to-End Process Flow

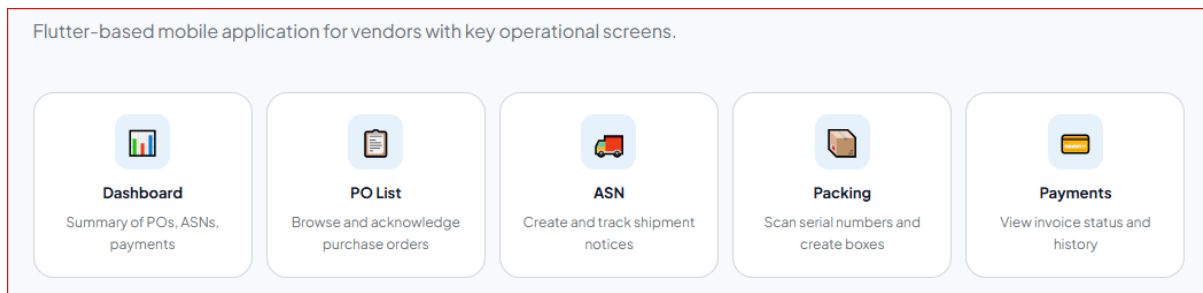


#	Step	Details
1	SAP PO Created	Procurement creates PO via ME21N in SAP
2	Portal PO Visible	PO pushed to portal via API/IDoc; visible on dashboard
3	Vendor Acknowledgement	Vendor accepts, rejects, or modifies the PO
4	ASN Creation	Vendor creates ASN with shipment details
5	Box + Serial Mapping	Outer/inner boxes created; serial numbers mapped
6	Label Printing	QR labels printed via PDF or Zebra printer
7	Dispatch	Goods dispatched to buyer location
8	GRN in SAP	Goods Receipt Note created in SAP FI/MM
9	Invoice	Invoice generated and submitted
10	Payment	Payment processed; vendor views status on portal

6. Mobile App Design

The mobile application will be built using Flutter, providing a cross-platform experience for both Android and iOS devices.

Flutter-based mobile application for vendors with key operational screens.



6.1 Key Features

- Scan serial numbers using device camera
- Create ASN from mobile
- Create boxes and manage packaging hierarchy
- Print labels via Bluetooth-connected Zebra printers



- View payment status and invoice history

6.2 App Screens

Screen	Description
Dashboard	Summary KPIs: Open POs, Pending ASNs, Payment status
PO List	Browse, filter, and acknowledge purchase orders
ASN	Create and track advance shipment notifications
Packing	Scan serial numbers, create boxes, print labels
Payment	View invoice status, payment history, and dues

7. Implementation Roadmap

Phase	Duration	Focus Area	Deliverables
Phase 1	6 – 8 Weeks	Foundation	PO Integration, Vendor Dashboard, Acknowledgement, Core Authentication
Phase 2	6 – 8 Weeks	Logistics	ASN Management, Box & Serial Creation, Label Printing
Phase 3	4 – 6 Weeks	Finance & Admin	Payment Visibility, User Management, RBAC
Phase 4	4 Weeks	Mobile App	Flutter App: all screens, Bluetooth label printing, serial scanning

High-Level Cost Estimation

Component	Effort Level
Backend Development	Medium
Frontend Development	Medium
Mobile App (Flutter)	Medium
SAP Integration (CPI)	High

8. Final Deliverables Summary

The following artefacts have been produced as part of this engagement:

- Complete Business Requirement Document (BRD)
- Functional Specification Document (FSD) — 8 Modules
- UI Wireframes for all key screens

- REST API Specification (SAP ↔ Vendor Portal)
- Database Schema — Simplified ER Model
- End-to-End Process Flow (10 steps)
- Mobile App Design and Screen List (Flutter)
- 4-Phase Implementation Roadmap with duration estimates
- High-Level Cost Estimation by component